DSA Tutorial - 5

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Topic: Implement Stack and Queue using arrays in C.

INPUT : STACK USING ARRAY -

```
#include <stdio.h>
#define n 5

int S[n + 1];
int top = 0;

void Stack_full() {
    printf("Stack is full\n");
}

void Stack_empty() {
    printf("Stack is empty\n");
}

void Push(int item, int S[]) {
    if (top >= n) {
        Stack_full();
        return;
    }
    top = top + 1;
    S[top] = item;
    printf("%d pushed onto stack\n", item);
}

int Pop(int S[]) {
    if (top <= 0) {
        Stack_empty();
        return -1;
    }
    int item = S[top];
    top = top - 1;
    return item;
}
```

```
void Stack_display(int S[]) {
    if (top <= 0) {
        Stack_empty();
        return;
    }
    int counter = top;
    while (counter > 0) {
        printf("%d\n", S[counter]);
        counter = counter - 1;
    }
}

int main() {
    int item;
    int ch;

    while (1) {
        printf("\n1. Push\n2. Pop\n3. Display\n4. Exit\nEnter a choice: ");
        scanf("%d", &ch);

        switch (ch) {
            case 1:
                printf("Enter Value: ");
                scanf("%d", &item);
                Push(item, S);
                break;
            case 2:
                item = Pop(S);
                if (item != -1) {
                    printf("Popped element: %d\n", item);
                }
                break;
            case 3:
                printf("Stack elements (Top to Bottom):\n");
                Stack_display(S);
                break;
            case 4:
                printf("Exiting....\n");
                return 0;
            default:
```

```
                printf("Invalid choice! Try again.\n"); }
    }
}
```

Queue Using Array -

```
#include <stdio.h>
#define MAX 5

int queue[MAX];
int front = -1 ;
int back = -1 ;

void enqueue(int value) {
    if (back == MAX - 1) {
        printf("Queue is Full!\n");
    } else {
        if (front == -1) front = 0;
        queue[++back] = value;
        printf("%d inserted\n", value);
    }
}

void dequeue(){
    if(front == -1 || front > back)
        printf("Queue is Empty\n");
    else{
    printf("Deleted: %d", queue[front++]); }
}
void display(){
    if(front == -1 || front > back)
        printf("Queue is Empty\n");
    else{
    for (int i = front; i <= back; i++) {
        printf("%d ", queue[i]);
    }
    printf("\n"); }
}

int main(){
    int value;
    int ch;
```

```
    while (1) {
        printf("\n1. Enqueue\n2. Display\n3. Dequeue\nEnter a choice: ");
        scanf("%d", &ch);

        switch (ch) {
            case 1:
                printf("Enter Value: ");
                scanf("%d",&value);
                enqueue(value);
                break;
            case 2:
                display();
                break;
            case 3:
                dequeue();
                break;
            case 4 :
                printf("Exiting....");
                return 0;
            default:
                printf("Invalid choice! Try again.\n");
        }
    }
}
```

OUTPUT:

STACK-

QUEUE-

Output

```
1. Enqueue
2. Display
3. Dequeue
Enter a choice: 1
Enter Value: 123
123 inserted

1. Enqueue
2. Display
3. Dequeue
Enter a choice: 1
Enter Value: 456
456 inserted

1. Enqueue
2. Display
3. Dequeue
Enter a choice: 2
123 456

1. Enqueue
2. Display
3. Dequeue
Enter a choice: 3
Deleted: 123
```

Output

```
1. Push
2. Pop
3. Display
4. Exit
Enter a choice: 1
Enter Value: 123
123 pushed onto stack

1. Push
2. Pop
3. Display
4. Exit
Enter a choice: 1
Enter Value: 456
456 pushed onto stack

1. Push
2. Pop
3. Display
4. Exit
Enter a choice: 3
Stack elements (Top to Bottom):
456
123
```